Practical 5 code

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Device Group Messaging"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="30dp" />

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your name"
        android:inputType="text" />

    <EditText
        android:id="@+id/etGroup"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter group name"
        android:inputType="text"
        android:layout_marginTop="15dp" />

    <Button
        android:id="@+id/btnJoin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="JOIN GROUP"
        android:layout_marginTop="20dp" />

</LinearLayout>
```

MainActivity.java

```
package com.example.groupmessaging;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   EditText etName, etGroup;
   Button btnJoin;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);

       // Connect Java variables with XML views
       etName = findViewById(R.id.etName);
       etGroup = findViewById(R.id.etGroup);
       btnJoin = findViewById(R.id.btnJoin);

       // Join group button
       btnJoin.setOnClickListener(v -> {

           String name = etName.getText().toString().trim();
           String group = etGroup.getText().toString().trim();

           // Check if fields are empty
           if (name.isEmpty() || group.isEmpty()) {

               Toast.makeText(MainActivity.this,
                       "Please enter name and group name",
                       Toast.LENGTH_SHORT).show();

           } else {

               // Open ChatActivity
               Intent intent = new Intent(
                       MainActivity.this,
                       chatActivity.class
               );

               // Send name and group to next screen
               intent.putExtra("name", name);
```

```
                intent.putExtra("group", group);

                startActivity(intent);
            }
        });
    }
}
```

chatActivity.java

```
package com.example.groupmessaging;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;

import com.google.firebase.database.DataSnapshot;
import com.google.firebase.database.DatabaseError;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.google.firebase.database.ValueEventListener;

public class chatActivity extends AppCompatActivity {

    TextView tvGroup, tvMessages;
    EditText etMessage;
    Button btnSend;

    String name, group;

    DatabaseReference databaseReference;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_chat);

        // Connect Java code with XML views
        tvGroup = findViewById(R.id.tvGroup);
        tvMessages = findViewById(R.id.tvMessages);
        etMessage = findViewById(R.id.etMessage);
        btnSend = findViewById(R.id.btnSend);

        // Get name and group from MainActivity
        name = getIntent().getStringExtra("name");
        group = getIntent().getStringExtra("group");
```

```
        tvGroup.setText("Group: " + group);

        // Firebase path for this group
        databaseReference = FirebaseDatabase.getInstance()
                .getReference("groups")
                .child(group);

        // Send message
        btnSend.setOnClickListener(v -> sendMessage());

        // Receive messages
        receiveMessages();
    }

    private void sendMessage() {

        String text = etMessage.getText().toString().trim();

        if (!text.isEmpty()) {

            String id = databaseReference.push().getKey();

            Message message = new Message(name, text);

            if (id != null) {
                databaseReference.child(id).setValue(message);
            }

            etMessage.setText("");
        }
    }

    private void receiveMessages() {

        databaseReference.addValueEventListener(
                new ValueEventListener() {

                    @Override
                    public void onDataChange(@NonNull DataSnapshot snapshot)
{

                        String allMessages = "";

                        for (DataSnapshot dataSnapshot :
                                snapshot.getChildren()) {

                            Message message =
                                    dataSnapshot.getValue(Message.class);

                            if (message != null) {
                                allMessages += message.sender + ": "
                                        + message.text + "\n\n";
                            }
```

```
                        }

                        tvMessages.setText(allMessages);
                    }

                    @Override
                    public void onCancelled(
                            @NonNull DatabaseError error) {
                    }
                });
    }
}
```

Message.java

```
package com.example.groupmessaging;

public class Message {

    public String sender;
    public String text;

    // Empty constructor required by Firebase
    public Message() {
    }

    // Constructor for creating a message
    public Message(String sender, String text) {
        this.sender = sender;
        this.text = text;
    }
}
```

Activity_chat.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="12dp">

    <!-- Group Name -->
    <TextView
        android:id="@+id/tvGroup"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Group Chat"
```

```
        android:textSize="22sp"
        android:textStyle="bold"
        android:gravity="center"
        android:padding="10dp" />

    <!-- Messages -->
    <TextView
        android:id="@+id/tvMessages"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:text="No messages yet"
        android:textSize="17sp"
        android:padding="10dp"
        android:background="#EEEEEE" />

    <!-- Bottom message area -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_marginTop="10dp">

        <EditText
            android:id="@+id/etMessage"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:hint="Type message"
            android:inputType="text" />

        <Button
            android:id="@+id/btnSend"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="SEND" />

    </LinearLayout>

</LinearLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <!-- Internet permission for Firebase -->
    <uses-permission android:name="android.permission.INTERNET" />

    <application
        android:allowBackup="true"
```

```
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.GroupMessaging">

        <!-- Chat screen -->
        <activity
            android:name=".chatActivity"
            android:exported="false" />

        <!-- Main screen -->
        <activity
            android:name=".MainActivity"
            android:exported="true"
            android:windowSoftInputMode="adjustResize">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>

        </activity>

    </application>

</manifest>
```

Build.gradle.kts (:app)

```
plugins {
    alias(libs.plugins.android.application)
    alias(libs.plugins.google.services)
}

android {
    namespace = "com.example.groupmessaging"
    compileSdk {
        version = release(36) {
            minorApiLevel = 1
        }
    }

    defaultConfig {
        applicationId = "com.example.groupmessaging"
        minSdk = 24
        targetSdk = 36
        versionCode = 1
        versionName = "1.0"
```

```
        testInstrumentationRunner =
"androidx.test.runner.AndroidJUnitRunner"
    }

    buildTypes {
        release {
            optimization {
                enable = false
            }
        }
    }
    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
    }
}

dependencies {
    implementation(libs.activity.ktx)
    implementation(libs.appcompat)
    implementation(libs.constraintlayout)
    implementation(libs.material)

    // Firebase
    implementation(platform("com.google.firebase:firebase-bom:34.18.0"))
    implementation("com.google.firebase:firebase-database")

    testImplementation(libs.junit)
    androidTestImplementation(libs.espresso.core)
    androidTestImplementation(libs.ext.junit)
}
```

Goggle-services.json

This will be different for each user

Libs.versions.toml

```
[versions]
agp = "9.2.1"
googleServices = "4.4.4"
junit = "4.13.2"
junitVersion = "1.3.0"
espressoCore = "3.7.0"
appcompat = "1.8.0"
material = "1.14.0"
activityKtx = "1.13.0"
constraintlayout = "2.2.2"

[libraries]
```

```
junit = { group = "junit", name = "junit", version.ref = "junit" }
ext-junit = { group = "androidx.test.ext", name = "junit", version.ref =
"junitVersion" }
espresso-core = { group = "androidx.test.espresso", name = "espresso-core",
version.ref = "espressoCore" }
appcompat = { group = "androidx.appcompat", name = "appcompat", version.ref
= "appcompat" }
material = { group = "com.google.android.material", name = "material",
version.ref = "material" }
activity-ktx = { group = "androidx.activity", name = "activity-ktx",
version.ref = "activityKtx" }
constraintlayout = { group = "androidx.constraintlayout", name =
"constraintlayout" }

[plugins]
android-application = { id = "com.android.application", version.ref = "agp"
}
google-services = { id = "com.google.gms.google-services", version.ref =
"googleServices" }
```

Build.gradle.kts (GroupMessaging)

```
// Top-level build file where you can add configuration options common to
all sub-projects/modules.
plugins {
    alias(libs.plugins.android.application) apply false
}
```